INCH-POUND

MIL-S-52881B
INTERIM AMENDMENT 1(ME)
30 September 1994

MILITARY SPECIFICATION

SERVICING UNIT, COMPRESSED GAS, MOBILE

This interim amendment is approved for use within the US Army Belvoir Research, Development, and Engineering Center, Department of the Army, with MIL-S-52881B, dated 11 May 1989.

PAGE 1

2.1.1, under SPECIFICATIONS, FEDERAL add:

"A-A-52428 - Reflector, Indicating Clearance."
"BB-A-1034 - Compressed Air, Breathing."

PAGE 2

Under SPECIFICATIONS, MILITARY, delete "MIL-R-12144" and "MIL-G-20241" in their entirety, and add:

"MIL-H-17672 - Hydraulic Fluid, Petroleum, Inhibited."
"MIL-O-27210 - Oxygen, Aviator's Breathing, Liquid and Gas."
"MIL-V-62038 - Vehicles, Wheeled: Preparation For Shipment and Storage Of."

Under STANDARDS, MILITARY, delete "MIL-STD-281" in its entirety.

PAGE 3

Under STANDARDS, MILITARY, delete "MS35387" and "MS51331" in their entirety, and add:

"MIL-STD-1330 - Cleaning and Testing of Shipboard Oxygen, Nitrogen, and Hydrogen Gas Piping Systems.
"MIL-STD-1366 - Transportability Criteria."
"MIL-STD-1622 - Cleaning of Shipboard Compressed Air Systems."

PAGES 4 and 5

2.1.2, delete "THE MILITARY TRAFFIC MANAGEMENT COMMAND", in its entirety.

Under "DEPARTMENT OF THE NAVY", delete "Naval Experimental Diving Unit Instruction 7121A" in its entirety.

AMSC N/A FSC 3655

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

PAGE 8

3.3.1.1, line 5, after "MIL-S-81733", delete "or chromate tape conforming to MIL-G-20241".

PAGE 9

3.5.1, lines 1 and 2, after "eight" delete "DOT 3AA standard". After "cylinders," delete conforming to RR-C-901,".

3.6.4, line 4, after "accordance with "delete MTMC/TEA pamphlet 70-1" and substitute "MIL-STD-1366".

PAGE 10

3.8, delete "e" in its entirety.

PAGE 11

3.10.1, line 10, delete "3.14.3.2" and substitute "6.14".

3.10.3, line 1, after "purifier(s)" add "filter(s)". Line 5, after "ASTM S30400" add "or S31600".

3.10.3.1, lines 3 and 4, delete the last sentence and substitute: "All fittings shall be of ASTM S30400 or S31600 material, and double ferrule type."

PAGE 12

3.10.3.2.1, line 4, after "ASTM S30400" add "or S31600".

3.10.3.2.3, line 2, after "ASTM S30400" add "or S31600".

3.10.3.2.4, line 9, after "ASTM S30400" add "or S31600". Line 12, delete: "The valve handle . . ."On" position."

3.10.3.3, change title to read, "Filter(s) and purifier(s)." Lines 2 and 3, delete "in figure 2 for type I servicing units and as shown".

Add the following sentence at the end of the paragraph: "An inline 90 micron particulate filter shall be incorporated as shown in figure 2 for the type I servicing unit."

3.10.3.4, line 2, delete "purifier cylinder" and substitute "90 micron filter".

PAGE 13

3.10.3.5, lines 1 thru 4, delete the sentence "The regulator shall . . . in the supply cylinder manifold.".

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:37Z
Check the source to verify that this is the current version before use.

3.10.3.6, lines 3 and 4, after "100 psi" add ", and 25 psi," and after "increments" add "respectively."

3.10.3.8, line 6, after "ASTM S30400" add "or S31600". Lines 11 thru 14, delete sentence "Flexible wire braided ASTM S30400 . . FED-STD-H28 (connection No. 870).", and add:

"Flexible stainless steel, Series 300, metal-wrapped hose charging leads (at least 15 feet in length) and yoke-type filling adapters shall be provided for connecting to Haskel pump connections for both air and oxygen servicing sides."

Line 15, delete "ASTM S30400 material." and substitute "chrome plated brass.".

3.10.3.9, line 4, delete "purifier(s)" and substitute "90 micron filter".

PAGE 14

3.10.3.10, line 3, after "DOT 3AA" add ", conforming to RR-C-901,".

At the end of the paragraph add: "Cylinder valve material shall be of chrome plated brass or stainless steel, preferred.".

3.10.4, line 7, delete "Department of the Navy's NAVSEA 0994-LP-001-9010", and substitute "MIL-STD-1472".

3.10.5, line 2, delete "wit" and substitute "with".

TABLE I

Page 16, under "VMRS 13-002-014 Lining, Rear Brake", delete lines 1 thru 4 and substitute

"- Lining shall be in accordance with DOT FMVSS for the specified braking system."

Under "VMRS 13-006-008 Plug, Master Cylinder Filler", delete lines 4 and 5 and substitute:

"- Provide surge actuator including the sealed filter cap that has an internal rolling diaphragm to trap air."

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

Page 17, under "VMRS 16 Suspension", line 18, add: "Full compression is defined to be the point where the suspension system can no longer be compressed. Rebound is defined to be the vertical distance from full compression to where the suspension system ceased to oscillate."

Page 18, under "VMRS 17-001 Tire, Pneumatic (continued)", delete lines 1 and 2 in their entirety.

Under "VMRS 18-003-001 Wheel Assembly (Rim & Wheel), Disc" line 3, delete "No. X-45249".

Page 19, under "VMRS 34-0003-092", line 2 of the title, delete "(MIL-C-1348612)" and substitute "(MIL-C-13486/12)".

Page 20, under "VMRS 34-004-0026 Reflector, Red (Side)", line 1, delete "MIL-R-12144, MS35387" and substitute "A-A-52428".

Page 22, under "VMRS 59-004 Drawbar", delete lines 3 and 4 in their entirety.

Page 24, under "VMRS 78-002 Fender Assembly, Rear", "VMRS 78-009 Deflector Assembly, Gravel", line 8, after "antisail type" delete "and interchangeable with MS51331." Lines 11 and 12, delete in their entirety.

PAGE 26

3.13, lines 1 and 2, delete "Department of the Navy NAVSEA 0994-LP-001-9010" and substitute "MIL-L-17672".

PAGE 28

3.14.3, delete in its entirety and substitute:

"3.14.3 Cleaning and sampling requirements for compressed air/oxygen. All parts and systems for use with the type I or type II sets shall be thoroughly cleaned and an analysis conducted in accordance with the requirements of MIL-D-16791 and MIL-STD-1300 or MIL-STD-1622. Compressed air sampling requirements shall be in accordance with BB-A-1034, source II, grade A. Oxygen sampling requirements shall be in accordance with MIL-O-27210, type II."

3.15, line 1, before "Data plates shall be applied" add "Two". At the end of the paragraph add: "One plate would be for all the overall system and the other would be for a manifold servicing unit."

PAGE 29

3.17, under "Description" for item 1, delete "6000 psi" and substitute "5000 psi", and for item 2, delete "2265 psi" and substitute "5000 psi".

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:37Z
Check the source to verify that this is the current version before use.

PAGE 33

4.4.2, line 2, delete "4.6.2.8" and substitute "4.6.2.9".

PAGE 34

4.6.1.1, a.(1), delete "NEDUINST 7121A", and substitute "MIL-STD-1300 or MIL-STD-1622". b.(2) and (3), delete in their entirety.

PAGE 35

TABLE III, under "Quality conformance", for "System Cleanliness", delete "X" and substitute "-".

Under "First Article", for "Environmental" delete "See 6.2" and substitute "X".

Add the following tests:

First Article	Quality conformance	Test	Test para	Reqmt para
X	-	Production System Cleanliness	4.6.2.12	3.14.3 thru 6.14
X	-	Human factors	4.6.2.10	3.7, 3.10.3.2.4
X	-	Safety	4.6.2.11	3.8
X	X	Inspection of packaging	4.7	Section 5

PAGE 37

4.6.2.3.2, a., line 1, after "the maximum" add "load".

PAGE 38

4.6.2.5, line 1, delete "Prior to oxygen cleaning,".

Line 5, before "Charge at least" add "Simultaneously". After "SCUBA cylinders," add "at a rate of 400 psi per minute,".

4.6.2.5.2, line 1, after "less cylinders" add "cylinder valves, relief valves, and pressure gages,".

Line 3, after "for the oxygen systems." add: "Servicing units components can be either individually tested or as a system.".

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:37Z
Check the source to verify that this is the current version before use.

PAGE 39

4.6.2.5.3, line 5, delete "NAVSEA 0994-LP-001-9010, chapter 5" and add "accordance with MIL-STD-1300 and MIL-STD-1622.".

Add the following new paragraphs:

"4.6.2.5.3.1 System cleanliness. The contractor shall meet the life support system cleanliness requirements, as specified in 3.14.3 and 6.14.

"4.6.2.5.3.2 Oxygen portion of system. An oxygen sampling test shall be conducted on the oxygen portion of the type II set using known oxygen values meeting the baseline requirements of MIL-O-27210, type II, table I as the input to the manifold servicing unit. A working pressurized sample shall be extracted at the outlet and analyzed against the baseline requirements of MIL-O-27210, type II, table I. Certification shall be provided that the input/output oxygen samples, when analyzed by one of the methods in MIL-O-27210 (4.8.4), meets the requirements of MIL-O-27210, type II, table I. Failure to provide this certification is cause for rejection.

"4.6.2.5.3.3 Compressed air portion of the system. A compressed air sampling test shall be conducted on the air portion of the type I/type II sets using known compressed air values meeting the requirements of BB-A-1034, source II, grade A, as the input the manifold servicing unit. A working pressurized sample shall be extracted at the outlet and analyzed against the baseline requirements of BB-A-1034, source II, grade A. Certification shall be provided that the input/output compressed air samples meets the baseline requirements of BB-A-1034, source II, grade A, tables I and II. Failure to provide this certification is cause for rejection."

PAGE 40

4.6.2.7 and 4.6.2.7.1, move both paragraphs to after 4.6.2.6.1.2.4.

PAGE 41

4.6.2.8.2, line 3, after "operated as in 4.6.2.5" add ", but with a flow rate of 200 psi per minute."

4.6.2.8.3, line 7, after "After the rain exposure" add "of thirty minutes".

After 4.6.2.11, insert new paragraph:

"4.6.2.12 Production system cleanliness. Certification shall be provided that each production unit has been cleaned and the analysis has been conducted as required by MIL-STD-1300 or MIL-STD-1622."

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:37Z
Check the source to verify that this is the current version before use.

PAGE 42

5.1.1, line 10, delete "MIL-STD-281, level A," and substitute "MIL-V-62038".

6.1, line 2, after "remote field sites.", add: "In addition, type I servicing units can be used to support surface supplied, lightweight diving operations to depths of 60 feet of seawater."

PAGE 47

Figure 2, delete and substitute new figure 2.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 3655-A127

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

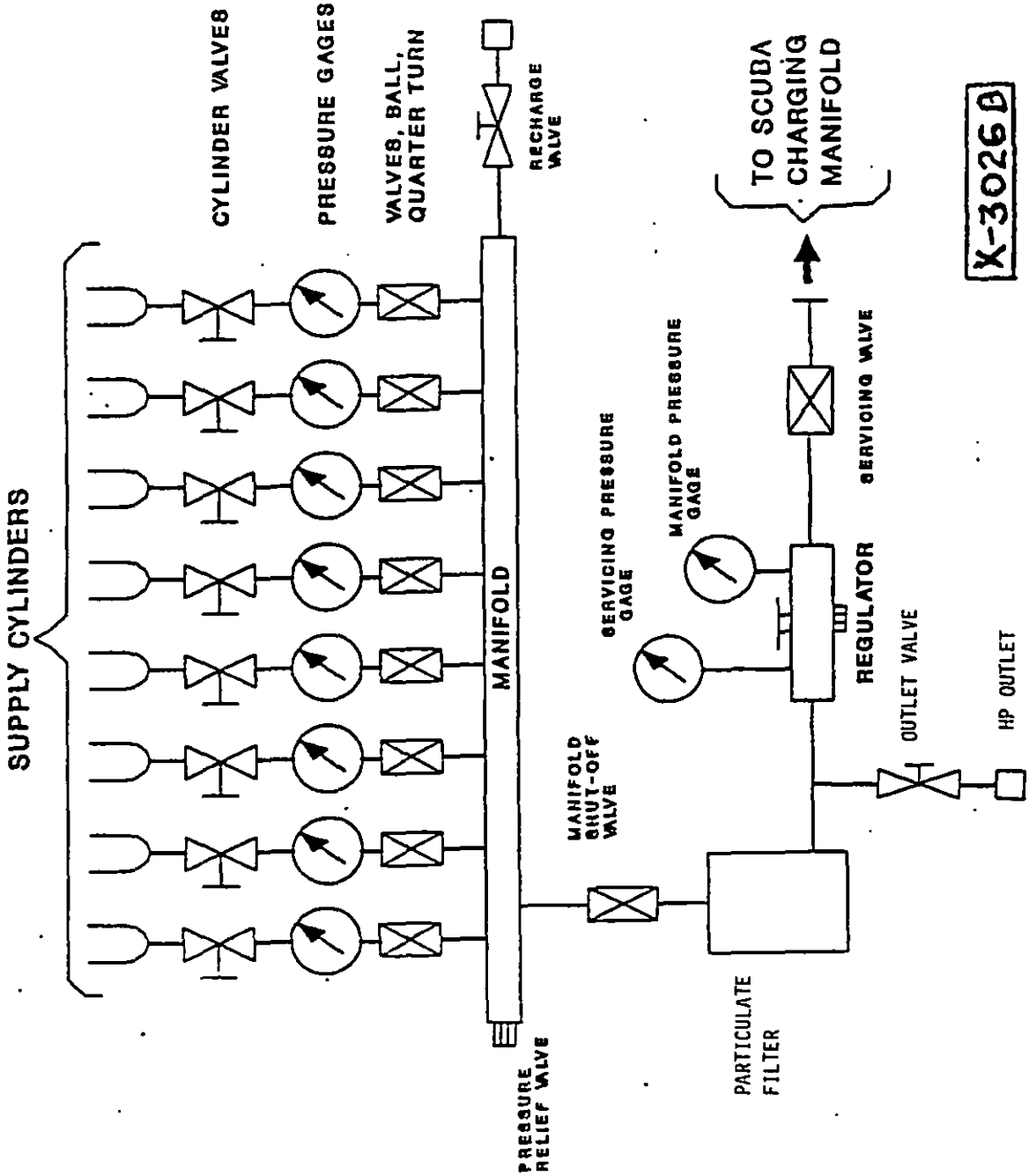


FIGURE 2. FLOW DIAGRAM, TYPE 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.